

HW 6

CSE 331

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Due: **Saturday** November 9th, 2024 11:00pm

Task 1 – The House That Back Built

1. Use forward reasoning to fill in the missing assertions (strongest postconditions).

Solution:

```
{ { y > 7 and z > 1 } }  
x = 3n * y + 1n;  
{ { x = 3y + 1 and y > 7 and z > 1 } }  
y = y - 7n;  
{ { x = 3y + 1 and y + 7 > 7 and z > 1 } } → { { x = 3y + 1 and y > 0 and z > 1 } }  
z = z * y;  
{ { x = 3y + 1 and y > 0 and z/y > 1 } }  
{ { x < 3z + 24 } }
```

We can see that this last assertion implies the stated postcondition $x < 3z + 24$ as follows:

$z > y$	since $z/y > 1$ and $y > 0$
$x = 3y + 1$	
$< 3z + 1$	since $y < z$
$< (3z + 1) + 23$	adding 23 to $3z + 1$ is bigger than $3z + 1$
$< 3z + 24$	

2. Use backward reasoning to fill in the missing assertions (strongest postconditions).

Solution:

```
{ { z > 0 } }  
{ { -z ≤ z } }  
x = -z;  
{ { x - 1 ≤ z - 1 } } → { { x ≤ z } }  
z = z - 1n;  
{ { x - 1 ≤ z } }  
y = x - 1n;  
{ { y ≤ z } }
```

We can see that $z > 0$ implies the condition we need since:

$$\begin{array}{ll} -z < 0 & \text{since } z > 0 \\ < z & \text{since } z > 0 \\ \leq z & \end{array}$$

3. Use forward reasoning, and prove by cases.

Solution:

```

{{ { x ≥ 1 and y = x2 }}
if (y < 20n) {
  {{ { x ≥ 1 and y = x2 and y < 20 }}
  y = y + 4n;
  {{ { x ≥ 1 and y - 4 = x2 and y - 4 < 20 }}
} else if (y < 28n) {
  {{ { x ≥ 1 and y = x2 and 20 ≤ y < 28 }}
  y = (y / x) + 1n;
  {{ { x ≥ 1 and x(y - 1) = x2 and 20 ≤ x(y - 1) < 28 }}
} else {
  {{ { x ≥ 1 and y = x2 and y ≥ 28 }}
  y = y / x;
  {{ { x ≥ 1 and xy = x2 and xy ≥ 28 }}
}
}
{{ { (x ≥ 1 and y - 4 = x2 and y - 4 < 20)
  or (x ≥ 1 and x(y - 1) = x2 and 20 ≤ x(y - 1) < 28)
  or (x ≥ 1 and xy = x2 and xy ≥ 28) }}
{{ { y > 4 }}

```

We'll prove by cases that $(x \geq 1 \text{ and } y - 4 = x^2 \text{ and } y - 4 < 20)$ or $(x \geq 1 \text{ and } x(y - 1) = x^2 \text{ and } 20 \leq x(y - 1) < 28)$ or $(x \geq 1 \text{ and } xy = x^2 \text{ and } xy \geq 28)$ implies the post condition $y > 4$:

- Assuming $x \geq 1$ and $y = x^2 + 4$ and $y < 24$:

$$\begin{array}{ll} x^2 \geq 1 & \text{since } x \geq 1 \\ y = x^2 + 4 & \\ \geq 1 + 4 & \text{since } x^2 \geq 1 \\ > 4 & \text{since } 1 + 4 = 5 > 4 \end{array}$$

- Assuming $(x \geq 1 \text{ and } x(y-1) = x^2 \text{ and } 20 \leq x(y-1) < 28)$:

$$\begin{array}{ll}
 x(y-1) = x^2 & \\
 y-1 = x & \text{since } x \geq 1 \\
 x(y-1) \geq 20 & \\
 (y-1)^2 \geq 20 & \text{since } y-1 = x \\
 y-1 \geq 2\sqrt{5} & \text{since } y-1 = x \geq 1 \\
 y \geq 2\sqrt{5} + 1 & \\
 > 4 & \text{since } 2\sqrt{5} + 1 > 4
 \end{array}$$

- Assuming $(x \geq 1 \text{ and } xy = x^2 \text{ and } xy \geq 28)$:

$$\begin{array}{ll}
 xy = x^2 & \\
 y = x & \text{since } x \geq 1 \\
 y^2 = xy & \\
 \geq 28 & \\
 y \geq \sqrt{28} & \text{since } y = x \geq 1 \\
 > 4 & \text{since } \sqrt{28} > 4
 \end{array}$$

Task 2– Hula-Loop

1. Prove that the invariant is true when we get to the top of the loop the first time.

Solution:

$$\begin{array}{ll}
 x_0 - 5y = x_0 - 0 & \text{since } y = 0 \\
 = x - 0 & \text{since } x = x_0 \\
 = x & \\
 x = x_0 & \\
 \geq 0 & \text{since } x_0 \geq 0
 \end{array}$$

2. Prove that, when we exit the loop, the postcondition holds.

Solution: The loop exits when $x < 5$, i.e. $-x > 5$ and $5 - x > 0$ by subtracting both sides by 5. Also, the initial condition $x_0 - 5y = x$ and $x \geq 0$ still holds.

$$\begin{array}{ll}
 5y = x_0 - x & \text{since } x_0 - 5y = x \\
 \leq x_0 & \text{since } x \geq 0 \\
 5(y + 1) = 5y + 5 & \\
 = x_0 - x + 5 & \text{since } x_0 - 5y = x \\
 = x_0 + (5 - x) & \\
 > x_0 & \text{since } 5 - x > 0
 \end{array}$$

3. Prove that the invariant is preserved by the body of the loop. Use either forward or backward reasoning (your choice) to reduce the body to an implication and then check that it holds.

Solution: We are proving using forward reasoning. Inside the while loop:

```

{{ Inv:  $x_0 - 5y = x$  and  $x \geq 0$  }}
while (x >= 5n) {
  {{  $x_0 - 5y = x$  and  $x \geq 5$  }}
  y = y + 1;
  {{  $x_0 - 5(y - 1) = x + 5$  and  $x \geq 5$  }}
  x = x - 5;
  {{  $x_0 - 5(y - 1) = x + 5$  and  $x + 5 \geq 0$  }}
  {{  $x_0 - 5y = x$  and  $x \geq 0$  }}
}

```

We want to prove that $\{\{ x_0 - 5(y - 1) = x + 5 \text{ and } x + 5 \geq 5 \}\}$ can imply to $\{\{ x_0 - 5y = x \text{ and } x \geq 0 \}\}$:

$$\begin{array}{ll}
 x_0 - 5(y - 1) = x + 5 & \\
 x_0 - 5y + 5 = x + 5 & \\
 x_0 - 5y = x & \text{subtract both sides by 5} \\
 x + 5 \geq 5 & \\
 x \geq 0 & \text{subtract both sides by 5}
 \end{array}$$

Hence, the invariant $x_0 - 5y = x$ and $x \geq 0$ is preserved in each iteration.

Task 3– Loop Dreams

The basic structure of the loop is as follows. You will fill in the missing pieces below.

```

{{ x > 0 }}
let v: bigint = 1n;
let w: bigint = 1n;
let y: bigint = 1n;
{{ Inv: (v - 1)2 < x and w = 2v - 1 and y = v2 }}
while (y < x){
    v = v + 1n;
    w = w + 2n;
    y = y + w;
}
{{ (v - 1)2 < x and x ≤ v2 }}

```

1. Fill in the initialization code above the loop. Then, prove that the invariant holds with your code.

Solution:

$(v - 1)^2 = (1 - 1)^2$	since $v = 1$
$= 0 < 1 = x$	since $x = 1$
$2v - 1 = 2 \cdot 1 - 1$	since $v = 1$
$= 1 = w$	since $w = 1$
$y = 1 = v^2$	since $v = 1$

2. Fill in the loop condition. Then, prove that the postcondition holds when the loop exits.

Solution: When the loop exits, $y \geq x$, and the invariant holds, i.e. $y = v^2$, so $y = v^2 \geq x$, and $(v - 1)^2 < x$ holds, i.e. the postcondition holds.

3. The first line of the body of the loop increases v by 1. Fill in the updates to w and y so that the invariant remains true with a v that is one larger.

Give the two lines of code. Then, use either forward or backward reasoning (your choice) to reduce to an implication and prove that it holds, showing that the invariant remains true.

Solution: See next page.

Solution: We are proving using forward reasoning. Inside the while loop:

```

{{ Inv:  $(v - 1)^2 < x$  and  $w = 2v - 1$  and  $y = v^2$  }}
while (y < x){
  v = v + 1n;
  {{  $(v - 2)^2 < x$  and  $w = 2(v - 1) - 1$  and  $y = (v - 1)^2$  }}
  → {{  $(v - 2)^2 < x$  and  $w = 2v - 3$  and  $y = (v - 1)^2$  }}
  w = w + 2n;
  {{  $(v - 2)^2 < x$  and  $w - 2 = 2v - 3$  and  $y = (v - 1)^2$  }}
  → {{  $(v - 2)^2 < x$  and  $w = 2v - 1$  and  $y = (v - 1)^2$  }}
  y = y + w;
  {{  $(v - 2)^2 < x$  and  $w = 2v - 1$  and  $y - w = (v - 1)^2$  }}
  {{  $(v - 1)^2 < x$  and  $w = 2v - 1$  and  $y = v^2$  }}
}

```

We want to prove that $\{\{ (v - 2)^2 < x$ and $w = 2v - 1$ and $y - w = (v - 1)^2 \}\}$ can imply to $\{\{ (v - 1)^2 < x$ and $w = 2v - 1$ and $y = v^2 \}\}$:

$$\begin{aligned}
 y - w &= (v - 1)^2 \\
 y - 2v + 1 &= (v - 1)^2 && \text{since } w = 2v - 1 \\
 y - 2v + 1 &= v^2 - 2v + 1 \\
 y &= v^2 && \text{add } 2v + 1 \text{ to both sides} \\
 x > y &= v^2 \\
 &> (v - 1)^2 && v^2 > (v - 1)^2
 \end{aligned}$$

Task 4– If It Ain’t Broke, Don’t Mix It

1. Prove that the invariant is true when we get to the top of the loop the first time.

Solution: Given $L = L_0$, $p = 0$, and $g = 0$ initially, by substituting the initial values into the invariant, we have: $p + \text{amt-purple}(L) = 0 + \text{amt-purple}(L_0) = \text{amt-purple}(L_0)$, and $g + \text{amt-green}(L) = 0 + \text{amt-green}(L_0) = \text{amt-green}(L_0)$, so the invariant is true.

2. Prove that, when we exit the loop, the postcondition holds.

Solution: After the loop we know that $L = \text{nil}$, so by the definition of amt-purple and amt-green , we know that $\text{amt-purple}(L) = \text{amt-purple}(\text{nil}) = 0$ and $\text{amt-green}(L) = \text{amt-green}(\text{nil}) = 0$. By the invariant, we have:

$$\text{amt-purple}(L_0) = p + \text{amt-purple}(L) = p$$

$$\text{amt-green}(L_0) = g + \text{amt-green}(L) = g$$

3. Use forward reasoning to fill in each of the P_i 's above and backward reasoning to fill in each of the Q_i 's above. Note that $L \neq \text{nil}$ means that we can write $L = L.\text{hd} :: L.\text{tl}$ since “ $::$ ” is the only non-nil constructor.

Solution:

```

{{ L = L0 }}
let p: bigint = 0n;
let g: bigint = 0n;
{{ Inv: amt-purple(L0) = p + amt-purple(L) and amt-green(L0) = g + amt-green(L) }}
while (L.kind !== "nil") {
  if (L.hd.kind === "RED") {
    {{ P1: Inv and L.hd.kind = "RED" }}
    {{ Q1: amt-purple(L0) = (p + 0.5) + amt-purple(L.tl) and amt-green(L0) = (g + 0.5) + amt-green(L.tl) }}
    p = p + 0.5;
    g = g + 0.5;
  } else if (L.hd.kind === "YELLOW") {
    {{ P2: Inv and L.hd.kind = "YELLOW" }}
    {{ Q2: amt-purple(L0) = p + amt-purple(L.tl) and amt-green(L0) = (g + 0.5) + amt-green(L.tl) }}
    g = g + 0.5;
  } else { // "BLUE"
    {{ P3: Inv and L.hd.kind = "BLUE" }}
    {{ Q3: amt-purple(L0) = (p + 0.5) + amt-purple(L.tl) and amt-green(L0) = g + amt-green(L.tl) }}
    p = p + 0.5;
  }
  {{ amt-purple(L0) = p + amt-purple(L.tl) and amt-green(L0) = g + amt-green(L.tl) }}
  L = L.tl;
  {{ amt-purple(L0) = p + amt-purple(L) and amt-green(L0) = g + amt-green(L) }}
}
{{ p = amt-purple(L0) and g = amt-green(L0) }}
```

4. Prove that the invariant is preserved by the loop body by showing that each P_i implies each Q_i .

Solution:

Inv: $\text{amt-purple}(L_0) = p + \text{amt-purple}(L)$ and $\text{amt-green}(L_0) = g + \text{amt-green}(L)$

P_1 : Inv and $L.\text{hd.kind} = \text{"RED"}$

Q_1 : $\text{amt-purple}(L_0) = (p + 0.5) + \text{amt-purple}(L.tl)$ and $\text{amt-green}(L_0) = (g + 0.5) + \text{amt-green}(L.tl)$

P_2 : Inv and $L.\text{hd.kind} = \text{"YELLOW"}$

Q_2 : $\text{amt-purple}(L_0) = p + \text{amt-purple}(L.tl)$ and $\text{amt-green}(L_0) = (g + 0.5) + \text{amt-green}(L.tl)$

P_3 : Inv and $L.\text{hd.kind} = \text{"BLUE"}$

Q_3 : $\text{amt-purple}(L_0) = (p + 0.5) + \text{amt-purple}(L.tl)$ and $\text{amt-green}(L_0) = g + \text{amt-green}(L.tl)$

• $P_1 \rightarrow Q_1$:

$\text{amt-purple}(L_0) = p + \text{amt-purple}(L)$	Inv
$= p + 0.5 + \text{amt-purple}(L.tl)$	$L.\text{hd.kind} = \text{"RED"}$, def of amt-purple
$\text{amt-green}(L_0) = p + \text{amt-green}(L)$	Inv
$= p + 0.5 + \text{amt-green}(L.tl)$	$L.\text{hd.kind} = \text{"RED"}$, def of amt-green

• $P_2 \rightarrow Q_2$:

$\text{amt-purple}(L_0) = p + \text{amt-purple}(L)$	Inv
$= p + \text{amt-purple}(L.tl)$	$L.\text{hd.kind} = \text{"YELLOW"}$, def of amt-purple
$\text{amt-green}(L_0) = p + \text{amt-green}(L)$	Inv
$= p + 0.5 + \text{amt-green}(L.tl)$	$L.\text{hd.kind} = \text{"YELLOW"}$, def of amt-green

• $P_3 \rightarrow Q_3$:

$\text{amt-purple}(L_0) = p + \text{amt-purple}(L)$	Inv
$= p + 0.5 + \text{amt-purple}(L.tl)$	$L.\text{hd.kind} = \text{"BLUE"}$, def of amt-purple
$\text{amt-green}(L_0) = p + \text{amt-green}(L)$	Inv
$= p + \text{amt-green}(L.tl)$	$L.\text{hd.kind} = \text{"BLUE"}$, def of amt-green

Thus, the invariant is preserved by the loop body.

Task 5 - Barking Up the Wrong Three

```

{{ L = L0 }}
let a: bigint = 0n;
let b: boolean = true;
{{ Inv: non-zeros(L0) = a + non-zeros(L) and even(L0) = (b = even(L)) }}
while (L.kind != "nil") {
  \ \ if L.hd === 0, do nothing
  if (L.hd === 1) {
    a = a + 1n;
    b = !b;
  } else if (L.hd === 2) {
    a = a + 1n;
  }
  L = L.tl;
}
{{ a = non-zeros(L0) and b = even(L0) }}

```

- a. Fill in the initialization code above the loop. Then, prove that the invariant holds with your code.

Solution:

$\text{non-zeros}(L_0) = a + \text{non-zeros}(L)$	Inv
$= 0 + \text{non-zeros}(L)$	$a = 0$
$= \text{non-zeros}(L_0)$	$L = L_0$

For $\text{even}(L_0) = (b = \text{even}(L))$, given that $b = \text{true}$ and $L = L_0$, we have $(b = \text{even}(L)) = (\text{true} = \text{even}(L)) = \text{even}(L)$. Since $L = L_0$, we have $\text{even}(L) = \text{even}(L_0)$, so $\text{even}(L_0) = (b = \text{even}(L))$ holds. Now we have proved that the invariant holds.

- b. Prove that the postcondition holds when the loop exits.

Solution: When the loop exits, $L = \text{nil}$ and the invariant tells us that

$$\text{non-zeros}(L_0) = a + \text{non-zeros}(L) \text{ and } \text{even}(L_0) = (b = \text{even}(L)).$$

Since $L = \text{nil}$, we have $\text{non-zeros}(L) = \text{non-zeros}(\text{nil}) = 0$, so the postcondition follows:

$$a = \text{non-zeros}(L_0) \quad \text{and} \quad b = \text{even}(L_0).$$

- c. Fill in the missing code in the body of the loop so that the invariant is preserved when L moves forward to the next element of the list.

Then, use either forward or backward reasoning (or both) to reduce the correctness of the loop body to implication(s) and prove that they hold.

Hint: Note that, if b and c are booleans, then “not $b = c$ ” is the same as “ $b = \text{not } c$ ”. Both expressions are true exactly when the values of b and c are different (one is true and one is false).

Solution: Inside the while loop:

```

{{ Inv: non-zeros( $L_0$ ) =  $a$  + non-zeros( $L$ ) and even( $L_0$ ) = ( $b$  = even( $L$ )) }}
while (L.kind !== "nil") {
  \ \ if L.hd === 0, do nothing
  if (L.hd === 1) {
    {{ non-zeros( $L_0$ ) =  $a$  + non-zeros( $L$ ) and even( $L_0$ ) = ( $b$  = even( $L$ )) and L.hd = 1 }}
    a = a + 1n;
    {{ non-zeros( $L_0$ ) =  $a - 1$  + non-zeros( $L$ ) and even( $L_0$ ) = ( $b$  = even( $L$ )) and L.hd = 1 }}
    b = !b;
    {{ non-zeros( $L_0$ ) =  $a - 1$  + non-zeros( $L$ ) and even( $L_0$ ) = (not  $b$  = even( $L$ )) and L.hd = 1 }}
  } else if (L.hd === 2) {
    {{ non-zeros( $L_0$ ) =  $a$  + non-zeros( $L$ ) and even( $L_0$ ) = ( $b$  = even( $L$ )) and L.hd = 2 }}
    a = a + 1n;
    {{ non-zeros( $L_0$ ) =  $a - 1$  + non-zeros( $L$ ) and even( $L_0$ ) = ( $b$  = even( $L$ )) and L.hd = 2 }}
  }
  {{  $P$  : (non-zeros( $L_0$ ) =  $a - 1$  + non-zeros( $L$ ) and even( $L_0$ ) = (not  $b$  = even( $L$ )) and L.hd = 1)
  or
  (non-zeros( $L_0$ ) =  $a - 1$  + non-zeros( $L$ ) and even( $L_0$ ) = ( $b$  = even( $L$ )) and L.hd = 2) }}
  {{  $Q$ : non-zeros( $L_0$ ) =  $a$  + non-zeros( $L.tl$ ) and even( $L_0$ ) = ( $b$  = even( $L.tl$ )) }}
  L = L.tl;
  {{ non-zeros( $L_0$ ) =  $a$  + non-zeros( $L$ ) and even( $L_0$ ) = ( $b$  = even( $L$ )) }}
}

```

P is gained from forward reasoning, and Q is gained from backward reasoning.

We want to prove P implies Q by analyzing each case in P :

- Case 1: $L.hd = 1$

$$\begin{aligned}
 \text{non-zeros}(L_0) &= a - 1 + \text{non-zeros}(L) \\
 &= a - 1 + (1 + \text{non-zeros}(L.tl)) && \text{def of non-zeros} \\
 &= a + \text{non-zeros}(L.tl) \\
 \text{even}(L_0) &= (\text{not } b = \text{even}(L)) \\
 &= (\text{not } b = \text{not even}(L.tl)) && \text{def of even} \\
 &= (b = \text{even}(L.tl))
 \end{aligned}$$

- Case 2: $L.hd = 2$

$$\begin{aligned}
 \text{non-zeros}(L_0) &= a - 1 + \text{non-zeros}(L) \\
 &= a - 1 + (1 + \text{non-zeros}(L.tl)) && \text{def of non-zeros} \\
 &= a + \text{non-zeros}(L.tl) \\
 \text{even}(L_0) &= (b = \text{even}(L)) \\
 &= (b = \text{even}(L.tl)) && \text{def of even}
 \end{aligned}$$

Both cases P imply Q , so we have proved the correctness of the loop body.